A.7 Log Book for Capstone Project

Mukesh Patel School of Technology Management & Engineering

Log Book for capstone project

Department: Information Technology

Program: Bachelor of Technology

Semester: VII

Project Title: RestockIQ: Demand Forecasting and Perishable Inventory Optimization for Food Retailers

Name of Faculty Mentor: Dr. Srikanth Pulipeti

Student Details

Std No.	Roll No	Name	Contact (per. email)
Student 1	A201	Yatin Patil	yatin.patil043@nmims.in
Student 2	A223	Chinmay Patil	chinmay.patil123@nmims.in
Student 3	A221	Soumy Raghuvanshi	soumy.raghuvanshi116@nmims.in

Group Members

Roll No.	Name	Sign (Students)	Grades (By Faculty Mentor)
A201	Yatin Patil		
A223	Chinmay Patil		
A221	Soumy Raghuvanshi		

Project Log Entries

Project Title: RestockIQ: Demand Forecasting and Perishable Inventory Optimization for Food Retailers

Week No.	Duration	Work Carried Out	Remarks by Faculty Mentor	Sign
1	13 Jul 2026 to 19 Jul 2026	Formed the project group under faculty mentor Dr. Srikanth Pulipeti. Initiated Literature Review & Requirement Finalization. Investigated operational challenges of daily food wastage and stockouts in artisan bakeries. Defined project aim, primary research questions, and structured the LaTeX review paper template.		
2	20 Jul 2026 to 26 Jul 2026	Conducted extensive literature survey covering 25+ papers on statistical time series (ARIMA, SARIMA, Prophet) and stochastic inventory theory (Newsvendor). Commenced Dataset Collection & Verification: sourced candidate POS transaction logs. Prepared the Interim Development Plan (IDP).		

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3	27 Jul 2026 to 02 Aug 2026	Acquired and ingested French artisan bakery POS dataset (2021–2022). Audited data quality, cleaned transaction logs, and mapped store closure days (Mondays and Wednesdays). Analyzed day-of-week sales variation and synthesized domain findings for requirements.		
4	03 Aug 2026 to 09 Aug 2026	Completed Literature Review & Requirement Finalization (100% completed). Finalized functional (FR1–FR7) and non-functional requirements. Prepared Review-I presentation deck. Presented Review-I to the faculty panel (Milestone: Review 1 Cleared at Week 4).		
5	10 Aug 2026 to 16 Aug 2026	Completed Dataset Collection & Verification (100% completed). Commenced Flutter App Development Setup: configured Flutter cross-platform repository, Material 3 design tokens, theme palette, and screen navigation architecture for store managers.		
6	17 Aug 2026 to 23 Aug 2026	Flutter App Development: built core mobile screens (dashboard overview, inventory cards, item list). Implemented baseline forecasting models (7-day rolling average, 4-week same-weekday average, seasonal naive) establishing benchmark lower bounds (WAPE 30.39%).		
7	24 Aug 2026 to 30 Aug 2026	Flutter App Development: built item detail view and manual stock check-in screens. Commenced Model Training (Prophet + XGBoost): implemented baseline Facebook Prophet model capturing weekly and yearly additive seasonality. Explored Google Gemini LLM for operational briefings.		
8	31 Aug 2026 to 06 Sep 2026	Advanced Flutter App Development to ~90% (waste alert UI, settings view). Explored GBDT regressors on Prophet residuals; observed extreme sparsity on low-volume individual SKU series. Prepared Review-II slide deck (G1-PPT-Review-II.pptx) with re-baselined Gantt chart.		
9	07 Sep 2026 to 13 Sep 2026	Presented academic Review-II to the departmental faculty review panel: demonstrated Flutter mobile application (95% complete) and initial forecasting baseline (30% model training progress). Received panel approval to proceed with backend API and cloud database integration.		
10	14 Sep 2026 to 20 Sep 2026	Completed Model Training (remaining 70%): overcame SKU sparsity by transitioning from individual tree models to a pooled gradient-boosted decision tree (LightGBM) trained across all items with lag and calendar features. Implemented adaptive fallback for sparse items.		

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11	21 Sep 2026 to 27 Sep 2026	Locked the production hybrid ML architecture. Completed remaining 5% of Flutter App Development (100% complete): finalized dynamic state management, real-time forecast binding, and waste alert cards. Integrated teammate pull requests and froze frontend interfaces.		
12	28 Sep 2026 to 04 Oct 2026	Provisioned and configured MongoDB Atlas cloud database cluster (restockiq). Designed collections for users, items, and batch forecasts. Conducted repository security audit, sanitized exposed credentials, and hardened .gitignore against secret leaks.		
13	05 Oct 2026 to 11 Oct 2026	Implemented Backend API, Restock Logic & LLM Narrator: built FastAPI REST service with JWT auth, waste mitigation discount tiers (HIGH 20%, MEDIUM 10%), and Google Gemini API (gemini-3-flash-preview) with zero-math prompt grounding. Completed System Integration (verify_e2e.py).		
14	12 Oct 2026 to 18 Oct 2026	Containerized FastAPI backend using Docker and deployed live to Render cloud hosting. Configured GitHub Actions CI/CD workflows, automated keepalive health pings, nightly batch forecasting scheduler, and reverse-proxy rate limiting.		
15	19 Oct 2026 to 26 Oct 2026	Testing & Evaluation: executed 38-fold weekly backtesting (Model A 26.06% pooled WAPE, Canonical Hybrid 22.51% summer folds), 200-iteration circular weather placebo test (weather rejected), and 68,883-unit Newsvendor simulation. Passed 55/55 test suite across all 5 tiers. Compiled Black Book and tagged v1.0-submission.		

Date: _____ Place: Shirpur	_____ **Dr. Srikanth Pulipeti** Faculty Mentor Department of Information Technology, MPSTME, Shirpur